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Threats to network security and Mitigating methods -DDoS Attack

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1 Abstract

The last few years technology "Internet" have become one of the major thing in our life it had changed the whole world from doing stuff physically were now we can do it virtually especially during the Covid Pandemic most of us are working from home or have business online and many other things. However the internet has also brought many vulnerabilities with it. Some network are not secured. This project is a demonstration of how to start securing a network for beginners showing how to and where to do it so that it is harder for the hacker to get access to the network and avoid unnecessary traffic inside the network.

This project will let you understand how the network work before/after security configuration like control access list have been applied and what each configuration exactly do to that specific IP address.

...

2 Project flow

Many things has been done during this project as shown below:

- Doing research about some DDoS attack and other security methods

- Come up with the perfect method to avoid unnecessary traffic.
- Build a small network with correct interfaces and IP addresses.
- Creating a webserver.
- Apply Password to the FTP server.
- Write The configuration carefully and correctly on a notepad first.
- Test the network after it is build to verify that everything is working fine before applying the ACL configuration.
- To control the devices how I want.

3 Introduction

Distributed Denial of Service (DDoS) and Denial of Service (DoS) attacks Nowadays are the major problem to the internet , We would be using an application then all of a sudden it shuts down, That's from DDoS and DoS attack. There are three types of DOS/DDOS attack SYN FLOOD, SMURF and FRAGGLE.

It was a 15 year old Canadian hacker Michael Calce known by "MAFIABOY" who is the first-ever to create a DoS attack that occurred in 2000 7th of February. Michael Calce arranged many of Dos attacks against few websites such as Amazon and eBay by using computers at multiple locations to make a traffic to the target computers and shut down their World Wide Web (WWW) sites[35].

Hacker sends numerous of SYN requests to flood a server with open connections. SYN consumes the server resources and make a huge traffic by sending enormous numbers of open TCP connections, making it difficult for the users who are connected to the server. SYN flood attack is easily to identify because the machine keeps crashing and becomes very slow.[7]

Another DDoS attack known as Smurf attacks is a network level attack. Smurf attacks is similar to ping floods, as it sends ICMP ECHO requests packets to the broadcast address of a network, So that the victim will be hit with many more packets. The attacker uses Smurf attack to let every machines on subnet respond to ECHO Request , after that the attacker forges the source address of ICMP ECHO with victim IP address. Fraggle

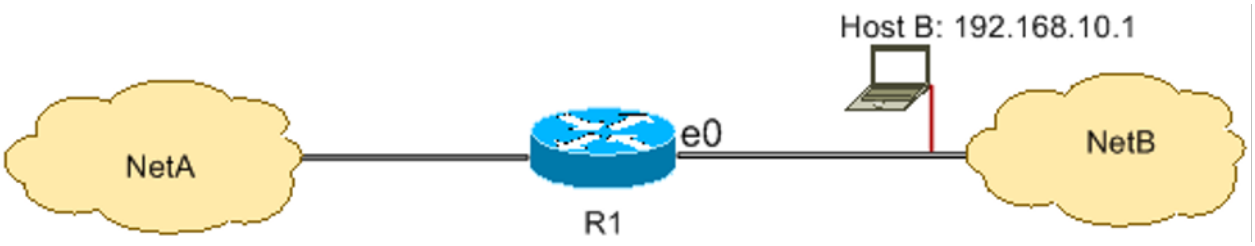
is a UDP packet magnification attack, it is similar to Smurf attack because fraggle uses the same style as the ICMP echo packets. Fraggle sends a huge amount UDP traffic to routers broadcast address and Smurf sends ICMP traffic to computer network using an IP broadcast address, but in the end they achieve the same goal.

Hackers use various DDoS Programs (TOOLS) to attack their victim. For the first tool Hacker uses panther program to flood their victim network with UDP packets, Land and LaTierra tool is used for opening TCP Connections and IP spoofing. Nemesey tool is used to generates random packets. Botnet gives you access to control many Compromised computers on the internet that hackers use Botnet to perfom a (DDoS) Distributed denial of service attack¿

- Panther
- Land and LaTieraa
- Nemesy
- Botnets
- blast

An access control list (ACL) has many access control enteries that identifies the traffic on the network and it also has the rules that gives you access or deny access to certain digital environment. For example gives you the ability to control the network traffic and reducing network attacks, So basically it is used to filter traffic as the rules we applied for incoming or outgoing network. Must setup

ACL must be applied to the router to deny any host from accessing the network. For example the photo below demonstrates the denied traffic going from Host B to Network A, while all other traffic are going through from Network B to Network A.

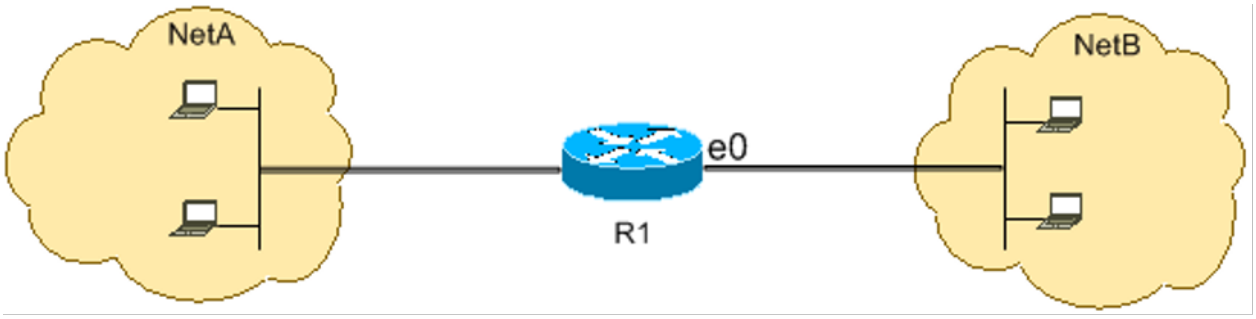


To deny Host B 192.168.10.1 harmful packets going through Router 1 from ethernet (0), the right way to do it is to apply these commands on Router 1.

- R1(config) Interface ethernet0
- R1(config-if) ip access-group 1 in
- R1(config-if) access-list 1 deny host 192.168.10.1

[15] [31]

Icmp pings Another example for Pings traffic (ICMP) let say hosts from Network B are trying to attack Network Aby pings. To deny all pings from network B to Net A and make ICMP sourced from Network A to Network B is permitted.

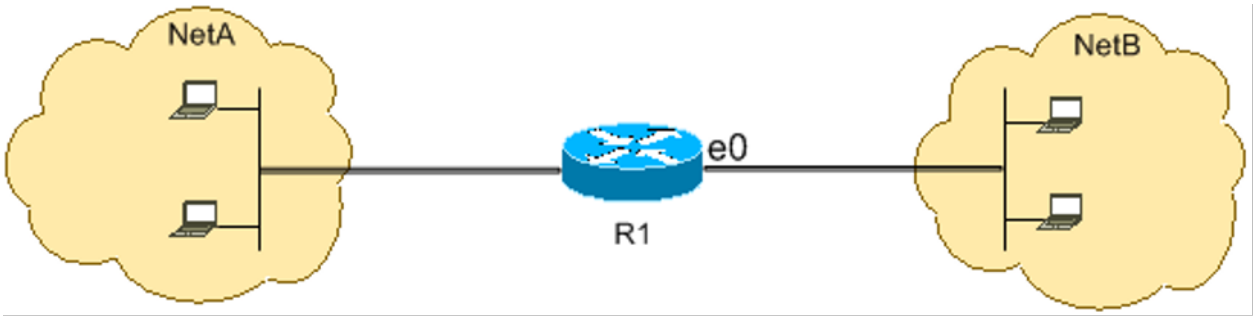


Configuration to allow ICMP sourced from Network A to Net B permitted and deny Network B pings to Network A.

- R1(config) interface ethernet 0
- R1(config-if) ip access-group 102 in
- R1(config-if) access-list 102 permit icmp any any echo-reply

[31]

TCP For TCP ACL 102 denies all datagram with an ACK/RST bit and any destination with port less then 1023 because the majority of ip service Ports is recognized and it utilize any value less then 102. For example the network below shows the TCP traffic from Net B to Net A is denied. So the purpose for adding ACL 102 command is to enable Hosts in Network A to start and maintain TCP Traffic session coming from Network B and deny Hosts in network B from attacking network A.



- R1(config) R1(config) interface ethernet 0
- R1(config-if) R1(config-if) ip access-group 102 in
- R1(config-if) aR1(config-if) access-list 102 permit tcp any any gt 1023 established

This configuration enables datagram to go through the interface when the datagram port value exceeds 1023. [17] [31]

4 Review of Literature

DDoS has a history that dates back in 1990, The WWW (web) was developing really quick that it has changed almost every aspect of our lives Almost most of the people back in the day the Web is their first choice to search for information or make online businesses or even make social relationships. Since many people started relying on the Web it increased the vulnerability against many different types of attacks on the computer network and hosts by compromising the Internet. One of the most dangerous threats to the web called DDoS attack **Distributed Denial of Service** it got well-recognized by hackers who aim to exploit the web weaknesses and those who work on defending any various attack to the web. [3]

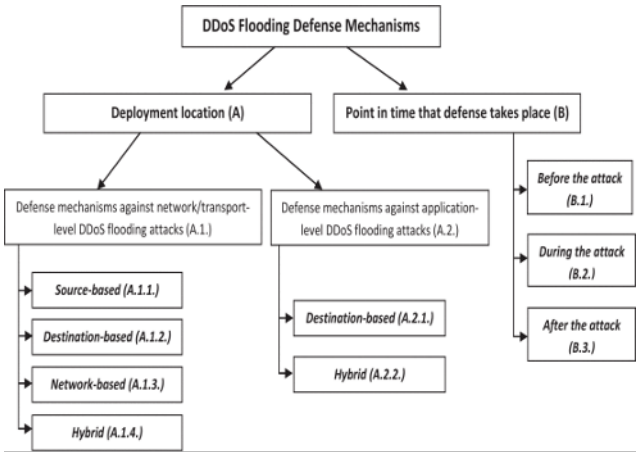
DDoS can affect computer networks in many ways, By that every Researcher have their own method to analyze and how to prevent their affected designed networks that got attacked by DDoS.

Rajani Muraleedharan et al.[24] published a study on 24 March 2006 where he has suggested a cross-layer security that can detect any DDoS attacks that are coming to the network and it prevents any crash that's happening at the network layers and data link. The cross layer security technique can be used to swaps the settings on the (OSI LAYER) Open Systems Interconnection to find the best solution, The authors plan is to have a QOS quality of service, high security and stable mobility when creating a cross-layer architecture.[37] Muraleedharan thinks that with this technique "cross-layer" is the best solution to give an accurate detection and also defends the network from DDoS attacks to other network layers, by obtaining an accurate result for detecting and preventing all incoming attacks to the network, than secure the network, this method will only allow the threats to interfere with the sensor physically.

In another study Anrien Bonguet et al (2017) [8]. mentioned that because of vulnerabilities that came with cloud features like resource sharing and multi tenancy, DDoS and DoS became the most dangerous threat to cloud based services, by discovering new various types of "XML DOS" and "HTTP DOS" threats such as **XML external Entity attack** targets application that parses the XML input and attack such as these may cause the disclosure and destruction of confidential information and also for DoS, server side request and forgery, port scanning from the machine that's hosting the parser, and other problems from the attack.[36] **HTTP flood** attacks a web server or application by using what appears to be legitimate HTTP GET or POST requests.[13] Both of these methods exhausts machine resources like Memory, Bandwidth, Computing Time. Bonguet made it clear that the hackers uses virtual machines or fake/damaged PC and laptops known as "bots" which can be controlled remotely. This can be done using BOTNET tool. Bonguet used 4 methods to mitigate this, detect the attack as soon as possible, secondly see what's the attack effecting, migrate the virtual machine that are under attack to a safe physical servers. Lastly guaranteeing network bandwidth that's how he prevent the attack from happening.

Taghavi et al[39] on the other mentioned that there is no solution for DDoS flooding attack after it is detected unless you remove the victim from the network and fix the issue by doing it manually. So Taghavi came up with

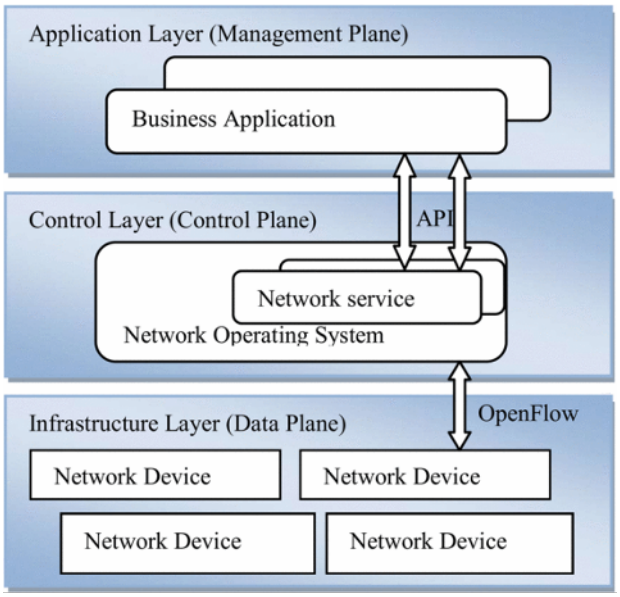
a new idea by reviewing and investigating the DDoS attack problem first then he analyzed the attacks and suggested a way by categorizing it in two criteria. His **first** criteria is where the defense mechanism is located and implemented then identified the mechanisms for the DDoS attack against transport level and the network by : destination, source, network and disturbed based, also he used the same technique destination based and hybrid based for the application level DDoS flood attack. On the **second** Criteria a DDoS attack must be detected at the same moment when the defense mechanism will be called upon and came up with two defensive options for application and network/transport level flooding attack. First is to avoid it before the attack which is called (attack prevention), Secondly must detect it during the attack and lastly identify and respond after the attack has happened. According to the authors they think that involving more nodes in the detection, prevention and response in order to implement hybrid and distributed defenses.[23, 40] The pictures below shows the defense mechanisms against Hybrid DDoS and DDoS flood.



Another study Cheung [9]analyzed and proposed countermeasures to the still existing attacks against DNS Domain name system service even though the proposed and existing countermeasures appeared to be useful, Cheung provided and recommended that more detailed analysis and have a deeper understanding of how it effect the service it will improve and gives an accurate security values. Cheung specified that many types of DoS attacks have been detected against DNS, from tampering with DNS settings to flooding attacks, these attacks prevent DNS requests from being served. He used do-

main name system security extensions (DNSSEC) it is the same as DNS that authenticates responses to DNL domain name lookups. It doesn't give any privacy safeguards for such lookups, but it does stop attackers from modifying or poisoning DNS request[4]. He also used peer to peer network technique that provides a more scalable alternative to fight against DNS flooding attacks. Peer to peer doesn't really depend on root servers and highest domain name system server for resolutions, and it could dynamically react to an increased amount of DNS requests for a domain by having additional nodes operate as servers.[11, 32]

Another study based on DDoS attack that uses SDN software for the detection of DDoS threats conducted by Myo Mynt Oo et al.[28] in August 23 (2018). First they structured the mechanisms in 3 layers for SDN "Infrastructure layer, Control Layer finally Application later". The infrastructure layer know by data plane it is established by basic network devices like switches, switches devices does not use higher systems administration capacities and just stop and sends the approaching packets as shown on the flow table that is arranged from the control panel and Software-Defined Networking "SDN" first standar utilizes open flow switch determination distributed by (ONF) Open Network Foundation. For the next part which is the Control plane (Control layer), it connects the two applications , the Management Plane and Data plane as shown in the picture below then the controllers are responsible for keeping a centralized view of network. It permits programs to manage any network protocol via open apis. Lastly is the third layer called Management plane "application Layer" it manages and secures the other two layers. Also it consist many applications that communicates their intended network behaviour and network needs to the SDN Controller both directly and programmatically.



J. Jiao et al. published a study in 2017 [20] in which they used Baidu data centers to test out their method and after that they detect DDoS attacks "TCP" by identifying random IP attacks and fixed IP attacks using Baidu cloud computing platform. First of all they used Packet sniffer to detect all incoming TCP flow then extract the TCP and IP header and each pair of IP addresses is divided by packet sniffer (Local hosts IP addresses, Remote host IP address) where they communicate with local hosts and every second, packets from remote IP to local IP are sent.

Jiao et al set a limit N on packets per second (PPS) as a threshold for IP-pair inbound traffic so that the system will issue an fixed source IP attack "FSIA" detection message if the IP pair PPS inbound surpasses "N". They claimed that this strategy will detect fixed source IP attack only. Researches also applied another threshold "T" on the overall amount of incoming traffic to locate Random source IP attack "RSIA", exclusively collecting packets between IP pairings that do not fulfill the initial the threshold "N".

In another study Rozekrans et al.[34] published a study on the 14th of February 2013 in which they came up with a defense technique called re-

sponse rate limiting (RRL) that reduces the number of generated responses by filtering and excluding those that exceeds a certain threshold. This is accomplished when a DNS server responds to a DNS request and stores the requestors IP address. If more than a certain number of responses are received, the server blocks requests from this IP address. The server decides how and where to respond to a request based on (IP), (NAME), and error-status to determine the problem. However, response rate limiting ignores the number of DNS requests received.

In detecting DDoS threats J. David et al.[12] proposed three mitigating methods in 2015 , First one is the **"flow aggregation"** is used to detect attacks based on flow. In addition they used **"fast entropy com"** to detect the DDoS threat as soon as possible. Furthermore to enhance detection accuracy, these researchers used an **"adjustable Threshold Algorithm"**.

On the other hand another study has been published by M. Alkasassbeh et al in 2016.[1]. In order to identify the sorts of threats named Smurf threats, User datagram protocol flooding , Web Flood, and SQL Injection DoS, The researchers used 3 techniques "Multilayer perceptron, Naive Bayes and lastly Random forest" which which they implemented in the database that specify the threats information and name . With **MLP** is simple, there is only one way for the traffic threats to the data , is the that the traffic coming from the transmitter and going to the receiver. It may take a long time for the weight of the network that start causing its predicted value is to have the same number similarity as the bias values.[22, 14] **Random Forest** will collect different amount of classification trees and for every tree is built based on different components of the general dataset. Trees that was found will be gathered and then put them into unlabeled category based on the number of aggregation prediction and beneath every tree an item will be established for them to vote and clarify the winner that has the most votes.

Naïve bayes technique is similar to the other two classifiers it applies its own rule and have a high accuracy of prediction of the features are conditionally independent[5].

The system that M. Alkasassbeh et al proposed 6 steps of collections datasets. First step is to gather as many network traffic as possible from intrusion detection system and then analyze it **(Gather and analyze)**,

delete all unnecessary files and duplicate files **(Preprossecing)**, the third step **(Extract features)** Analyzes the data gathered in the network flow and extracts the feature information. Fourth step (**(Statistical measurements)** they used statistical formula is used to determine the extra features. Fifth step they used **(Training Algorithm mitigating method)** to pick only one single model that shows and defines the values of the connections from a different models.

Then lastly for the sixth step apply **(the transfer function)** to every neurons network input to be able to identify its output signal. During the experiment they found out that MLP was the best method out of the three techniques in which it achieved the highest detecting accuracy rate (98.63 percent) Are these technique good against DDoS attacks? First the random forest isn't stable you have to be correct before structuring because if you make a slight change to data settings it can lead to a big change to a structure. On the other hand MLP examines its structure really well but it doesn't provide us the information about the functions that are being calculated on the structure. However every one of the features that are in Naïve Bayes are considered to be independent. So it is much harder of getting predictors that are independent in an actual life. It is impossible to use Naives bayes in real life MLP doesn't provide any with information and Random isn't stable. It is a bit risky using these methods because if anything goes wrong or change a small thing then we have to change the whole structure again which takes a long time to do.[33, 27, 30]

Another study found by M. Nagaratna et al.[26] in 2009 where they identified three mitigating methods against distributed denial of service threats, (Preventive Defense , source tracking and reactive solutions). the first part is to find a way to make the security level high and enhance it to avoid and resist any kind of threat from happening to computers or network. Based on roaming algorithm, it is used to secure by moving between the active server and among the few distributed homogeneous servers which will allow only the real users to see the information of server time and the what address is the new server carrying.

Furthermore they explained that the source tracking is used for tracking the attack sources to identify what type of attack is it and what does it to so that they know how to prevent the attack and there has been different approaches of identifying the internet protocol packets that stores the routing destination information inside them, in order to route them throughout the

internet. For the reactive solutions, where the Distributed denial of service defense mitigating method is intended for highlighting/determine the current threat that's going on and then respond to it as soon as possible which requires the Control of the attack packets flow in order to prevent the attack.

The researchers will now be using (EMDAF) encrypted marking based detection and filtering to improve the attack detection and prevent it. The EMDAF has "three" functions, the system gathers the user packets IP address and marking value then this system will have to confirm if the marking value is correct by sending pings to source IP address, so the firewall will create a key if it matches the server marking value but if the they do not match the server will consider it as an attack and cancel the request. That's how they filter the attack, so this system EMDAF targets only IP spoofed in three ways it marks the process which need to ingage with the routers in order to change the destination information into packets, then filters the process and lastly secure the transmission.

Another study published by O. A. Osanaiye [29] in 2015 about how to detect and mitigate IP spoofing DDoS attack in Cloud computing. It is similar to M. Nagaratna et al mitigate method but more efficient and up to date. Osanaiye also mentioned that IP spoofing is one of the dangerous threats out there in networking, so he came up with another plan to mitigate it, furthermore his first method was "OS fingerprinting" which specifies the operating system of a remote computer through internet, also the method should be active or passive "it can do it both way" to sniff any packets that are being sent across the internet between number of hosts and to filter all fake IP packets by matching operating system of traffic packets received[38]. between number of hosts and to filter all fake IP packets by matching operating system of traffic packets received. This Operating system "Fingerpring" most of the time it is used by Administrators to see if their network has an old OS, also used to detect and upgrade any weak OS to the latest version, it also detects hosts that carry viruses. Passive fingerprinting gathers the header information from the incoming packets uses either passive operating system fingerprint (POF) or use passive Os fingerpring for Internet Protocol tables or Ettercap. For the active fingerprint it sends crafted packets to a real source using one of the three tools Nmap, Xprobe2 and SinFP. Osanaiye used different tool called NMAP which is a mapping tool for network. NMAP tool has a scanning capability which sends 15 probes that contains TCP, UDP and ICMP to hosts ports, after that it generates all responses Ip headers and

examine it to know what Operating system its running on this is done by comparing the response that was detected to the operating system database that has been stored in. Passive operating system fingerprint technique is used to identify the host operating system by scanning their TCP and IP packet header information. This can be done by fingerprinting the hosts Syn packets that are connected to a server and identify the SYN, ACK responses that's coming out from the server. Osanaiye used both of these methods passive and active OS fingerprinting which proves that the incoming packet sources from the target host with high accuracy detection this is achieved by determining their operating system in a cloud computing system, with this plan the author think this plan will improve the performance by automating the process[2].

F. A. Barbhuiya et al [6]. published a study on the 6th of September 2012 about a detection mechanism for ICMP Based attacks, where the authors described different types of security problems and threats that has got to do with ICMP using active detection technique that captures " Internet Control Message Protocol" threats to the network as soon as possible and they mentioned that this method is so effective to a point where it is showing great number of results. This can be done without using any ICMP packets format modification or need to update all network devices, however their method requires just a few CPU and memory resources since their mechanism is a software based. Internet Control Message Protocol messages is divided into two separate groups, first is the Error message that gathers false or fake packets and alarm the user about the issue. Second is the Informational messages which is used to analyze and troubleshoot the issue after an ICMP attack. The hacker that uses ICMP has various ways to do it.

- **1.Destination Unreachable Messages** where the attacker use to flood their victim device with these messages.
- **2.Source Quench Messages** slows the victims device which they cannot do anything about it.
- **3.ime Exceed Messages** the attackers also use this technique to deny the victim from responding, it is similar to Source Quench.
- **4.Paramater Problem Messages** the attacker gather the Header

parameters messages that has been notified to the user and forge it to start their attacking process.

- **5.Photuris** Attacker uses this technique to send DoS attack by corrupting the LAN and WAN and internet communication.

- **6.Router Redirect** Router inform the user all the best routes to a specific destination, then attackers take their advantage to start their attacking process by giving the user a false route and apply (MiTM) Man in the Middle attack to it.

The authors used the sniffer software to detect any incoming network traffic and stores their IP packets in the database but before storing the IP packets, the sniffer has to identify if it is fake or not first, then if it is fake the software notifies the user about the issue and then the software delete it before it is stored into the database. This mechanism only detects ICMP error messages that are coming for the attacker. //Is it promising using a packet sniffer software? If you are an expert using the software then the answer will be yes, but what about the beginners? It is difficult for new users to understand how the software work and what each color represent because you might see two same packets but with different colors so the sniffer can be confusing sometimes. Not just that, also it captures any other packets that are not an actual network traffic, so it captures all sorts of packets. Sniffer are really useful but it is really hard to find a specific threat because I captures many packets.

L. Mutu et al. [25]published a study on 7th of December 2015 about how to mitigate UDP flood attacks using UDP actuating triggers, these triggers are applied on the switch it operates against any UDP attack, based on a certain rules they applied, so that the switch alerts the controller if anything was wrong and make the admin to act immediately. These triggers can also be used again TCP traffic attack. **What are the actuating triggers?**

- **1. Traffic Percentage Trigger** Traffic percentage trigger which is based on to collect the percentage in a certain time for UDP traffic attack going to other networks, For example if a host keep sending traffic time after time to only 1 destination of a network, the Trigger will consider that the host is using any DDoS or DoS attack software

such as Botnet, even though the host isn't using any of these software. The trigger job is to measure the percentage of UDP packets that are being sent to a certain destination in a period of time that's been applied. So that if the packets are being sent exceeds a certain time the attack is detected then switch should act against the attack as instructed by the controller. //

- **2. An Amplification Rate Trigger** Amplification Rate Trigger focuses on UDP message responses ratio, Sometimes UDP response can be bigger then then the request, because of that the trigger is applied so that if a certain network obtains a large amount of traffic more than expected in just a few seconds, the trigger will consider it as an amplification attack. The defender Analyzes the number of traffic that are going through the ports in which the amplification is being placed For example the Ne twork Time Protocol "NTP" where it can be used to match the network server time and the hosts time. The amplification attack appears on 123 port and also occurs on 53 port for the DNS where the attacker sends as many requests as possible, If the traffic reaches its destination within the time that the applied that's where the administrator commands the switch to start action against the attack.

The authors used traffic percentage trigger for the User Datagram Protocol traffic going on the network. Also used amplification Identifier for the incoming UDP traffic, they believe that this mitigating method is important and really good for addressing "UDP" problems. So does this solve the problem? No, UDP is unreliable because you cannot be sure if the data is delivered or not delivered after it was sent and also it doesn't use any error control to correct any error and its careless because UDP doesn't notify you if there is any attack or flow going on, so with UDP the routers are not verified, any problem could occur at any time but you will not know about it. Because of these kind of problems, in my project I will be digging deep to find a better solution for mitigating any kind of DDoS attack and stop and UDP flows.

5 Access control List

ACL plays a major role in networking instead of having firewalls, we apply ACL rules to allow or deny any packet that been sent from the other user (Destination). If a company wants to really secure their network is to configure their routers manually by applying ACL rules because They can decide who's allowed to access their network or not and to avoid unnecessary packets with that there will be less incoming traffic if they manage it correctly for example Wireshark is a software that detects all incoming packets or any unwanted files not just that some websites use fake IP address so with wireshark captures millions of packets in just few seconds so it is hard to see all of the captures one by one furthermore not many people use wireshark even the networking professors don't use this software, which means users that don't have any knowledge about networking or packets they wouldn't know what might happen to their data and we users and Companies we do not want that to happen so instead we just apply The ACL commands to make sure that any other network that doesn't have to do with the internet protocol (IP) that we applied wont passthrough, Also ACL are modifiable you can always go back and change or add entries on a specific directory but we must make sure it is the correct IP address and it is applied on the correct router interface otherwise it will be a major problem and some Ip address we allowed wont be able to access the network and many other things could happen. ACL access control list is used for letting specific hosts to have the right to access any file or let the packets to get from one network to other network, It is basically all about allowing or denying specific user from accessing the network. Security is Important Nowadays since some Hackers try to send large packets to the user network to harm your Pc or even get access to your files, But most of the times hackers target Big Companies to try to pull some information out of the company Database and there are Many ways for the hacker to get access and it can also be used for avoiding big incoming traffic as well not just for securing it. What most of Security use is configure ACL onto the router since its set by default, So the Admins has to do it Manually for each router. There are 2 types of ACL first is the Standard ACL and Extended ACLs,

The Standard ACL- Uses Internet Protocol version 4 addresses to scan Layer 3 which is the network layer, It allows you to allow the source IP address but you cannot control the destination address . To create a standard ACL you must choose a number from 1-99 or 1300-1999 or do you can create named

standard acl to make it easier. Where do you want to place the standard acl? if the standard acl is applied on the router that is close to the source network. It can affect other network with huge traffic, the traffic will be all around the network which don't have the acl. So the best way to avoid this situation is to apply the commands on the destination network.[19]

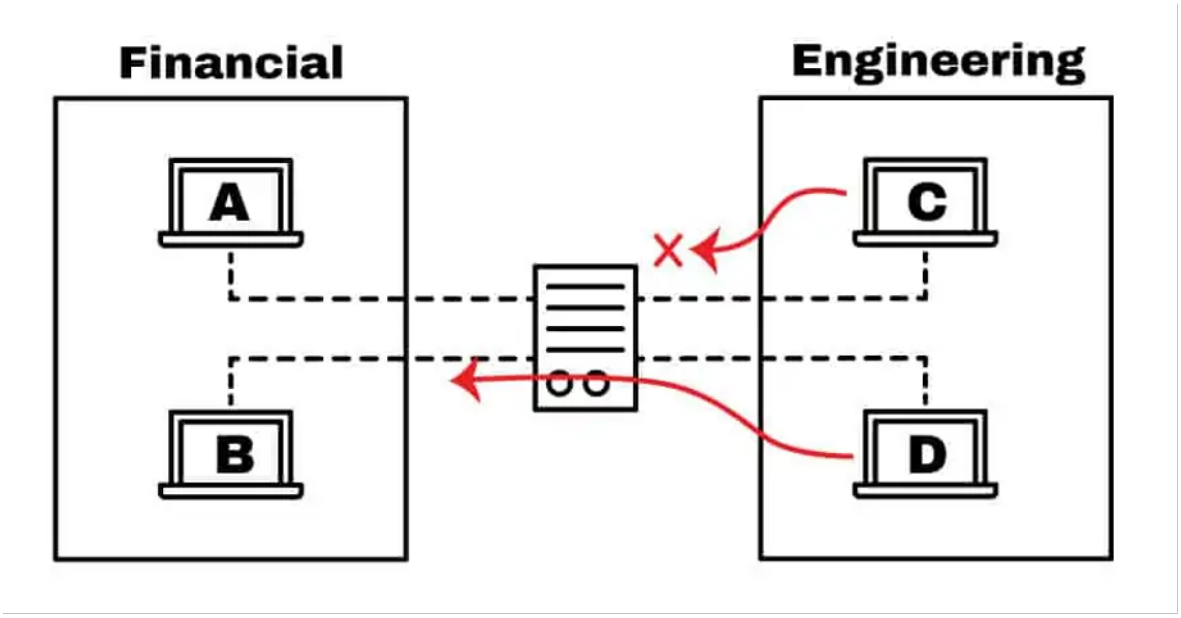
The Extended ACL- is the new and advanced version which can do the standard ACL job by filtering the layer 3 which is the network with Ipv4 addresses and scan the transport layer in layer 4 with Transmission Control Protocol, User Datagram Protocol USB Ports. With Extended ACL allows you to do manage the network however you want, which you can allow and block the IP address at once and let you control the destination of the source can do more things then just source and destination it even allows you to control based on User datagram protocol and Transmission Control Protocol port number and other port values. An extneded ACL is always viewed at ¡source¿ ¡destination¿. [10]

To create an extended list you must choose a number between 100 to 199 and 2000-2699 if you have many ACL. It can also be Named instead of number to make it more Understandable and which the acl belongs to. Where to apply Extended acl? it should be applied to router that is close to the users because the traffic will take some of the bandwidth until the destination even though it cannot proceed to the users, We do not want that to happen so we apply it on the closest router to the users.

One important thing about ACL commands, is that before placing it on the router you must first discuss it with the team if you are working as a group and plan it together, Also it must be written on notepad, Microsoft word or where ever you want. It will save you much more time and can edit it anytime, Once you apply it on the router it is hard to edit it especially if you already have other acl on the router then you have to edit the whole ACL because one acl command can affect the whole network, so you must be sure about each of the command and where it's place.

ACL is not a software or Network manager table or it is already applied within the router, It is all about configuration done by applying acl commands to the router. The commands isn't hard but what is hard is that you need to be focused and need to be placed carefully otherwise it going to mess the whole network and sometimes difficult to modify because sometimes if you change 1 thing then you have to do the same for others, So it is a bit difficult if you have a large network with bunch of commands anything could go wrong.

I will be discussing more about the ACL step by step as I am going through it.



6 Table: ACL Components

Sequence Number	Sequence Number is the numb the ACL, for example standard 1-99 and 100-199 for the exter
Access control list Name	Use ACL name rather then a nu to make it easier to you know one it belongs to
Remark	Remark is a command you car describe or identify where the belongs to or write whatever y want, Its your statement.
Statement	Statement allow or disallow an address going from a network another, by Using deny or perr the ACL
Network Protocol	Is the IP address ,UDP , TCP an ect, For example Deny ICMP....
Source	Let the source network to deny allow one or many IP address
Destination	Same thing with Destination, A you to deny or permit one or n address
Log	Allow some Users to save the l after the ACL is matched
EQ	EQ means equal to the port yo specifiy such as HTTPS OR HTT 443 and 80 port numbers

7 Why Access control list?

Since there are many Hackers out there trying to push traffic into one single network to shut it down, ACL let you avoid that by matching the incoming IP address with your IP address and if they do not match then the incoming traffic wont be allowed to go through the network and the same thing goes with ICMP and UDP.

Using access control list can improve and secure the network if you manage to do it correctly, It is the best way to deal with incoming traffic and it is manageable easily, it can do many thing such as:

- It capture traffic that you specified.
- Give priority to specific network users.
- Allows you to identify and control the traffic.
- Traffic limitation so not all traffic go through.
- Deny or permit users for accessing the network.

Most of big companies uses ACL to avoid the traffic in their network since they have many users, Without it the network would be corrupted, some users will have low latency which results to "LAG" slow internet connection, also the network might shutdown because it cannot handle big traffic. So that's where ACL comes in to manage the whole network and giving each user access and every traffic has its own direction where its supposed to go and avoid Unwanted and dangerous traffic that's coming from unknown websites or anywhere else. ACL makes the network invulnerable and really hard for attackers to get access to the network and it organizes the whole network so that every section has its own job, So that if one problem occurs in one of the ACL list only that ACL list will have the problem not the whole network.

The ACL happens to be very useful for advanced IT users, It can be little bit difficult for beginners but once you master it everything will be easy, It only requires some thinking, Time and especially Patience also it must be done on a notepad first so you can comeback to it later on and edit it, then transfer the whole configuration from the notepad to the router configuration mode. Should make sure that it's the right ACL configuration for the right router interface otherwise the network will be messed up. In most cases ACL

are done in groups “as a team” since it is a bit complicated especially when implementing it to big network there will be many ACL commands so one person cannot do all of it by himself, It is really difficult for one person to apply the whole configuration for big network so that’s why it is done as team and each one has his one section to add the ACL, Applying it as team will save time and gives you time to go through the commands and make sure it is right.

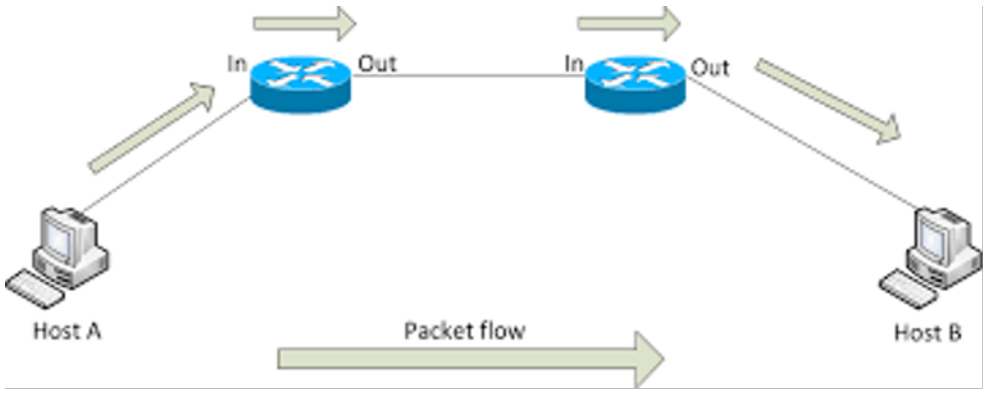
After applying the ACL the network would be safe and easier to manage and control the traffic and users, and Companies won’t be worried about unwanted traffic. So the hard part is done and now all the IT department has to do is to download security softwares on each laptop to maintain the security.

8 How does ACL Operates?

The ACL rules that were applied on the router will look at the packet that are coming through the inbound Interface and the flow in the router that has been caused by Relay which happen when the source and the destination IP address connects using other nodes, So the Router transfer the data info to many other devices. Furthermore the ACL would also verify the packets which are leaving the routers outbound interface because ACL can also be applied on the routers to In/outbound traffic.[21]

Before applying the in/outbound on the router we need to understand it first how it works, When you apply the Inbound it will capture and filter the packet before letting it go through the outbound interface, So basically it has to confirm and match the details we applied on the ACL to let it go through. The inbound can be really useful because it will save the router from having traffic before even getting into the router, also it saves the packet information if it is declined to pass through.

On the other hand Outbound is used for filtering the packets after they are inside the router and any other packets that are going out of the router is considered as outbound.



The Access control list that was configured within the router interface it will start following 4 steps to stop the traffic that has gone through the inbound interface of the router.

- First The original IP address version 4 is going to be extracted from the header of the packet by device.
- Then the router will verify if the Source IP address matches the access control entries in sequence order, starting from the above of the access control list.
- Thirdly when the IP address matches the ACL we applied it will then carry the instructions on whether packet are allowed to go through or not.
- Lastly when the IP addresses do not match with the instructions we applied on the router it will the discard the IP and not allow it to go through since it did not match the access control entries we configured.

The Access control entries usually an implied deny that wont give any access to any packets that are passing through because the ACE is automatically applied on the ACL and it is not visible. When configuring the access control list you should apply 1 permit statement other then that the traffic will not be allowed due to the access control entries. ACL can be really confusing for some people and that's why most of them don't pay attention to it, and it's time consuming as well and must do it carefully but in the end it is really important Companies to apply the ACL otherwise the whole network will be messed up and some users will have connection problem or the employers won't have access or the is too slow for them to serve the costumers such as the cashiers in grocery stores or restaurants.

9 What are DDoS or DoS attack type?

There are many ways for the hacker to attack and shut down their network, It can be done by flooding the network using these sources : **Internet Control Message Protocol (ICMP), transport protocol (TCP), User Datagram Protocol (UDP), Domain Name System (DNS), Dynamic Host Configuration Protocol (DHCP) and Address Spoofing.** These are the major sources that most of the hackers use to flood and break down the whole network.

9.1 What is ICMP attack and how does it work?

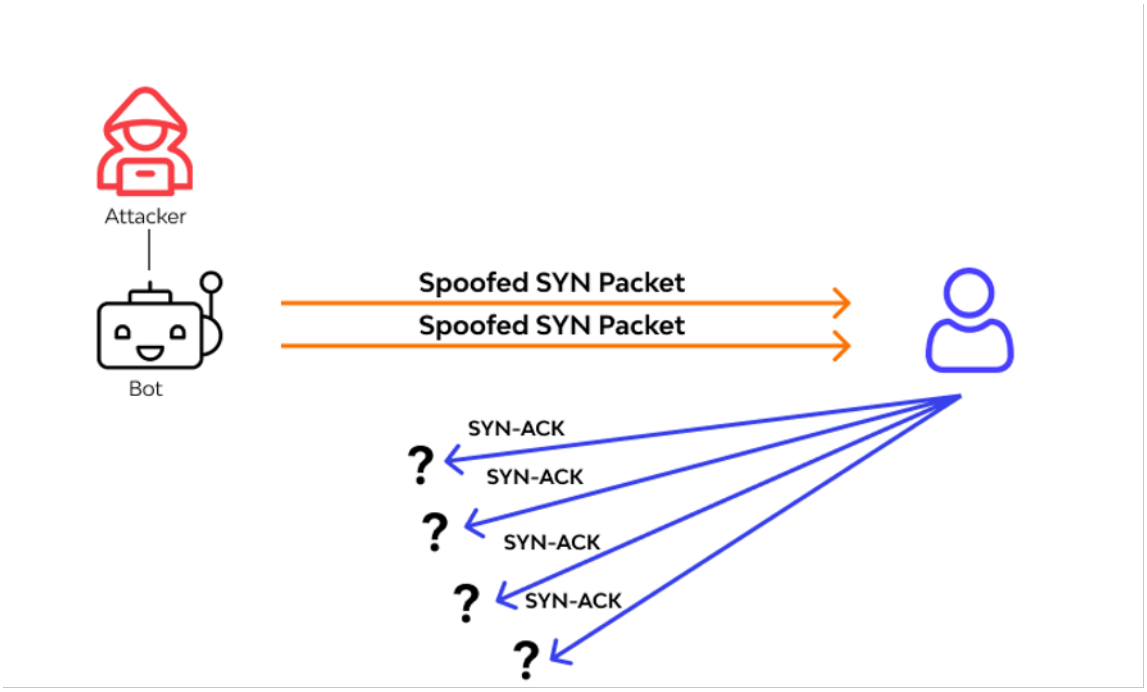
ICMP is an internet control message protocol that is used for Requesting messages and Receiving response from other devices which is called ping and that's how network admin sometimes do to see if there is any issue with the device. So now when it comes to attackers they take an advantage from it where they start using different software tools that allows them to use thousands of fake or virtual computers and start pinging the target after sending the request packet to the user then the router will have to send the same number of response packet as the request and it is done from both side the incoming requests and outgoing responses by this it will consume the router bandwidth because of the traffic going on and so the router won't handle it then it will shutdown So this is called a DDoS attack.[18]

It can also be done using the network devices that are not working or not being used

9.2 What is TCP attack and how does it work?

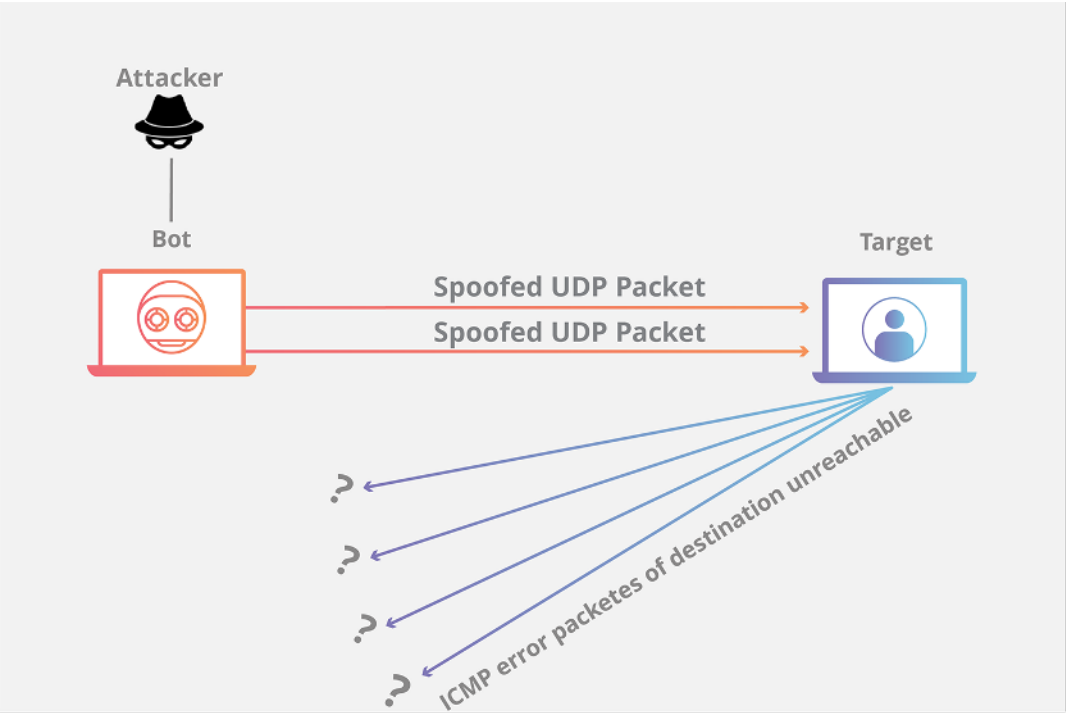
TCP is a transport protocol that is used to verify the network to fix or avoid expired packets, two identical packets, fake packets but the hackers use it to shutdown and reset the user device or website by Flooding Syn. So how does it work? The hacker has to check if the user device can listen to other devices then if the user device do not listen, so the hacker will start sending the TCP reset to the device and shut it down. So hackers actually use this method when the device is not aware (Listening) of what's happening. However if the device does listen to other communication then the hacker won't be able to send anything because the user device is aware of everything and it will not accept any spoofed Syn packet .

This picture below show's how the attacker use bot to send spoofed syn packets to access the user device and shutdown the network from the user device.[16]



9.3 What is UDP attack and how does it work?

UDP is User Datagram Protocol that all network devices has it, it is like an Identity card of the user detail but placed on the computer which specifies the IP address ect.... UDP attack is similar to TCP and ICMP attack where it sends thousands of UDP to any of the victim port. How does it work? When the hacker sends multipleUDP to the user by generating fake IP Address the victim system will then verify if the UDP that has been received matches the port or belong to the datagram, If it doesn't match then the system will send an ICMP reply to the attacker that the destination is unreachable since UDP is connectionless and the victim system keeps doing it for thousands of fake UDP. So that is how the attacker do to flood the system.



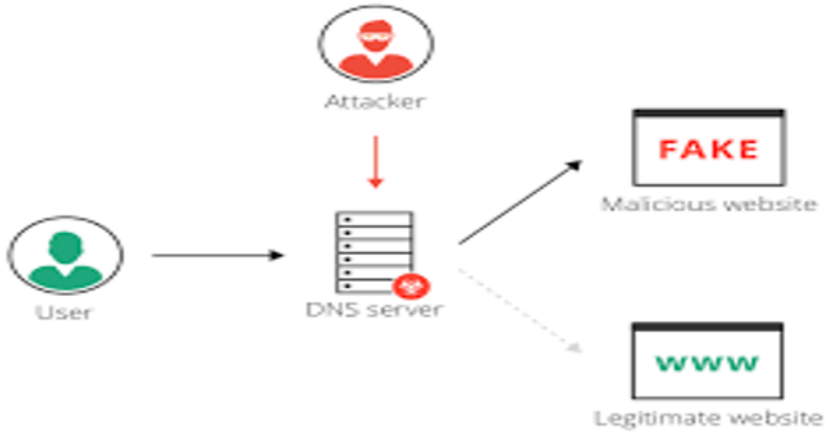
9.4 DNS attack and how it is done?

Domain name system "DNS" is a system that convert or translate the user name, website URL for example google.com or goal.com to IP addresses so that people can browse it using the IP address for such 8.8.8.8 address for google. So how does the hacker take advantage of this technique and use it as weapon to send bunch of traffic to the target network? the way the Hackers use DNS attack is to look for an open DNS servers to overload the victims network by DNS responses. First the attacker generates thousands of fake IP addresses "Ip spoofed address" then sends DNS look-up request to the server using a fake IP that matches one of the victim IP address after that the Server will then send all the responses to that specific victim. There are many more ways of DNS attack.

9.4.1 Such as DNS Hijacking where hackers:

Change the DNS IP address to the one that their own IP address so that the users information head to the attack IP address. It can also be done by

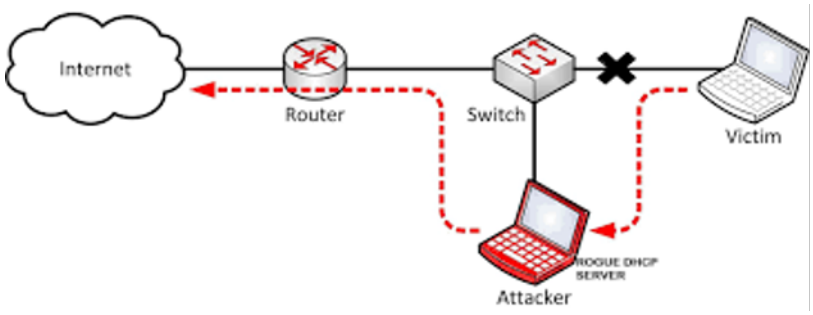
compromising the victim account and change their DNS server name for a server name that the attacker is using. Lastly hackers also change the router server so whenever a user logs in the router will then automatic sends the server to the users.



9.5 What is DHCP attack and how it is done?

(DHCP) Dynamic Host Configuration Protocol can be done on one router where the router assign different IP addresses Subnet mask default gateway to all the PC's that are connected to the DHCP server. How do the attackers use this as an advantage and try to overload the victim network and devices with traffic? It is similar to other type of DDoS attack but this one is done using DHCP server where the attacker send many of requests to the DHCP server to use all IP addresses. After that the DHCP do not have as many IP address as that has been requested from the attacker so this will lead to traffic and DDoS attack.

Other way the hackers take over the DHCP is called DHCP spoofing, where the hacker put their own dhcp server configuration for the network so that when the user's join the network they will be assigned with fake DNS that leads them to use the hacker machine default gateway. As you can see in the picture below the victim doesn't use the real switch that is connected to the internet but instead they are connected to the attacked device to get to the internet.



10 What are the four important steps before applying configuration?

The first step is to assign Privileged EXEC Mode on devices such as the switch and routers. There are two EXEC modes First one is the user exec where the user can only access ¿ prompt and apply basic configuration for example to see the status, like using telnet or maybe ping other devices, see the Ip address or the interfaces these kind of stuff. The user cannot do anything besides that Neither can restart the device. It is basically read-only for the user.

```
R1>?
Exec commands:
  <1-99>      Session number to resume
  connect     Open a terminal connection
  disable     Turn off privileged commands
  disconnect  Disconnect an existing network connection
  enable      Turn on privileged commands
  exit        Exit from the EXEC
  logout      Exit from the EXEC
  ping        Send echo messages
  resume      Resume an active network connection
  show        Show running system information
  ssh         Open a secure shell client connection
  telnet      Open a telnet connection
  terminal    Set terminal line parameters
  traceroute  Trace route to destination
```

The second one is the Privileged EXEC mode where it is mostly used by the administrator where it will allow you to use the User exec mode and the

configuration mode where you will be able to do whatever you want to the device such as applying or modifying the configuration, Naming the device so you have access to control the device unlike the user mode. After you enter the ¿ prompt then you must enable to access the configuration mode which is the prompt.

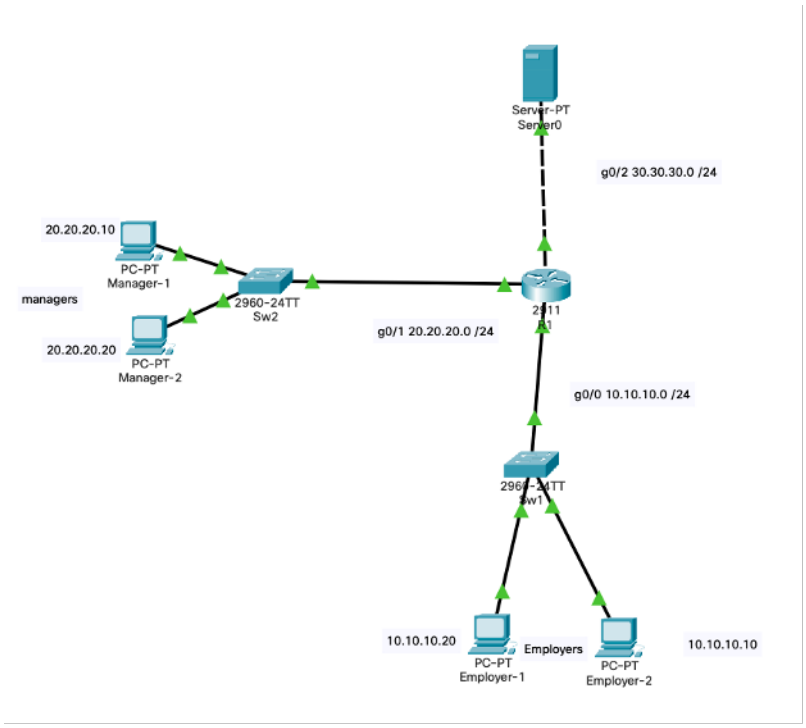
This must be done to the network devices to avoid all necessary access.

```
R1#
R1#
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#?
Configure commands:
  aaa                 Authentication, Authorization and Accounting.
  access-list         Add an access list entry
  banner              Define a login banner
  bba-group           Configure BBA Group
  boot                Modify system boot parameters
  cdp                 Global CDP configuration subcommands
  class-map           Configure Class Map
  clock               Configure time-of-day clock
  config-register     Define the configuration register
  crypto              Encryption module
  default             Set a command to its defaults
  do                  To run exec commands in config mode
  dot11               IEEE 802.11 config commands
  enable              Modify enable password parameters
  end                 Exit from configure mode
  exit                Exit from configure mode
  flow                Global Flow configuration subcommands
  hostname            Set system's network name
  interface           Select an interface to configure
  ip                  Global IP configuration subcommands
  ipv6                Global IPv6 configuration commands
  key                 Key management
```

11 Creating the lab on packet tracer:

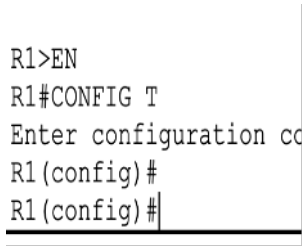
I will be creating a lab for small private company to demonstrate and give you a good idea about how the PC's can communicate with the server and each other without any problem and without ACL being applied, Having a big lab can confuse some people I'll try to make it as much easier as I can for the reader. In this section I will be using 2 switches, 1 router, 1 Internet Server and 4 PC's to make it easier for the reader to understand and have a good idea about it. To make it simpler for the reader is to give every area its own interface so for PC 0 and pc 1, I gave them g0/0 interface with 10.10.10.0 network and name that network Employers. For the pc's

3 and 4 I gave them G0/1 interface with 20.20.20.0 network and name it Managers. Lastly G0/2 network 30.30.30.0 for the Server. This is how the lab looks, I gave an example for a small company with default configuration being applied simple and easy to understand.



This network above do not have any ACL configuration on and no password applied to each device. As you can see it's not asking me to put any password in to be able to enter the configuration mode. I am able to access the devices without any issue.

Check the picture below.



```
SW1>en
SW1#config t
Enter configurati
SW1(config)#
```

- Let's try to ping from pc1 to other devices to check if they can communicate. All devices can ping each other.

```
C:\>ping 20.20.20.10

Pinging 20.20.20.10 with 32 bytes of data:

Request timed out.
Reply from 20.20.20.10: bytes=32 time<1ms TTL=127
Reply from 20.20.20.10: bytes=32 time<1ms TTL=127
Reply from 20.20.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 20.20.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 20.20.20.10

Pinging 20.20.20.10 with 32 bytes of data:

Request timed out.
Reply from 20.20.20.10: bytes=32 time<1ms TTL=127
Reply from 20.20.20.10: bytes=32 time<1ms TTL=127
Reply from 20.20.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 20.20.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- Let's try to access the FTP from pc 1. All PC's are able to access the FTP successfully

Note: The Username and Password I have given to the FTP is cisco.

Let's try to access the internet server. All PC'S are successful accessing the internet.

11.1 What we learnt from this experiment?

In this experiment we learnt that all the PC's are able to communicate with each other such as the Employers and the Managers. And the Employers are able to access the ftp. The network devices such as the switches and router

it doesn't have any password available. So with default configuration the network is not safe and hackers can attack this company so easily without any issue even if the password is configured for each device. Hackers have many solutions to attack the network. The main idea of this project is to manage this network and give the right access to the right devices and to avoid unnecessary IP address that are coming in, we want to keep this network private for the company only not for users outside the company. On the next stage I will be showing you how to manage it correctly step by step and see the difference between the two networks. After all that the network should be ready and safe for use and work perfectly without any problems.

12 Demonstration/Evaluation

In this section I will be demonstrating step by step on how and where to apply the access control list and see what's going to happen to the network after its being configured. First I want to allow the communication between the employers and the managers so that they can ping each other. Second is to allow the Employers to access the internet webserver. Thirdly disallow the Employers to use the File transfer protocol server "FTP". Finally disallow Employers network or any other network trying to access the server unless it is permitted for them to communicate with specified network.

The configuration must be done on the router. So on R1 type Ip access-list Extended "Name it to whatever you want" in my case I named it ControlEmployers as you can see on the picture below. Now you that's where all the magic happens. The most important part.

```
R1(config)#ip access-list extended ControlEmployers
R1(config-ext-nacl)#
```

1.The configuration starts now so first I want to allow any Employers network 10.10.10.0 to be able to access the Webserver. With this configuration I am allowing any device from the specific network to be able to access the internet server which is the employers "permit tcp 10.10.10.0 0.0.0.255 host 30.30.30.10 eq 80". So in this case I am permitting that specific network by using EQ "equivalent" and "80" is a http server port.

```
R1(config-ext-nacl)#permit tcp 10.10.10.0 0.0.0.255 host 30.30.30.10 eq
www
```

2. On the same access-list I want to allow the communication between the two network "Managers and the Employers" so that they can ping each and receive information about each other. The configuration for it is "permit ip 10.10.10.0 0.0.0.255 20.20.20.20 0.0.0.255" Which means permit any device between employers and managers IP addresses.

```
R1(config-ext-nacl)#permit ip 10.10.10.0 0.0.0.255 20.20.20.20
0.0.0.255
```

3.3. On the first step I allowed the 10.10.10.0 network to access the network. However I don't want the Employers to communicate with the server. So I am going to Deny the communication between the server and the Employers by**"deny ip 10.10.10.0 0.0.0.255 host 30.30.30.10"**

```
R1(config-ext-nacl)#deny ip 10.10.10.0 0.0.0.255 host 30.30.30.10
```

4. Step 4 I am going to deny The employers to access the FTP "deny udp 10.10.10.0 0.0.0.255 host 30.30.30.10 eq 53". 53 port stands for domain.

```
R1(config-ext-nacl)# deny udp 10.10.10.0 0.0.0.255 host 30.30.30.10 eq
domain
```

5. Step 5 is to deny the packets that are coming from the employers and its destination is from a different network. All I do is type "deny ip 10.10.10.0 0.0.0.255 any"

```
R1(config-ext-nacl)#deny ip 10.10.10.0 0.0.0.255 any
```

6. The final step is the most important part even though I applied the commands it will not work because I haven't placed them to the interface. So In this case I have to apply the configuration to the Employers nearest interface which is g0/0. **Go to interface g0/0 ¿ Ip access-group ControlEmployers in.** that's it everything is done. Keep in mind this is just a test to show you how it works so This configuration will not affect the managers network because they can do whatever they want.

```
R1(config-if)#int g0/0
R1(config-if)#ip access-group ControlEmployers in
```

This how it should look if you show IP access-lists after the configuration is done. On the next step I will be testing it and give you more knowledge about it

```
R1#show ip access-lists
Extended IP access list ControlEmployers
    10 permit tcp 10.10.10.0 0.0.0.255 host 30.30.30.10 eq www
    30 deny ip 10.10.10.0 0.0.0.255 host 30.30.30.10
    40 permit ip 10.10.10.0 0.0.0.255 20.20.20.0 0.0.0.255
    50 deny ip 10.10.10.0 0.0.0.255 any
    60 deny udp 10.10.10.0 0.0.0.255 host 30.30.30.10 eq domain
```

12.1 Why and How to apply password to each device?

12.1.1 Assign and enable Line console and Line V T Y to each device.

Line console is used for connecting the routers and switches through a console cable. If some devices on the network are connected through a console cable then you should enable Line con to each device that are connected by applying password for example if there is one port only then apply line console 0 but if you have more then one port then put how many ports you've got for example line console 2. Lastly put a password to that specific Console then enable it by login.

- Router(config) line console 0
- Router(config-line) password cisco
- Router(config-line) login

12.2 Line VTY:

securing virtual port such as telnet and SSH for example you are on a router and would like to access a different router or switch. So when you try to access different device through telnet or SSH it will ask you for the password by that you will not be able to access and manage the other device.

- Router(config) line vty 0 4
- Router(config-line) password cisco
- Router(config-line) login

The 0 4 which mean the device would only support up to five virtual connection at once.

12.3 Final step:

The final step is to encrypt the password to have a better security and make it hard for the hackers to find the password. It is only one line configuration where it actually hide the actual password and replace it with a random number for example when an unauthorized user get access to the device and show the device status it will show him a random password number.

Notice the difference before the configuration is applied and after it's applied. The password is showing on the first picture but now on the second one which has been replace with random number

```
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 password cisco
 login
!
line aux 0
!
line vty 0 4
 password cisco
 login
line vty 5 15
 password cisco
 login
!
!
!
end
```

```
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 password 7 0822455D0A16
 login
!
line aux 0
!
line vty 0 4
 password 7 0822455D0A16
 login
line vty 5 15
 password 7 0822455D0A16
 login
!
!
!
end
```

12.4 How to apply password to the devices

How to apply password to the devices

I will assign a password to each device So that when the hacker pass through the ACL and try to access the device it will ask him for a password. It is an additional security but it is important to keep the network safe incase anything happens Because companies and even Normal Users would like to secure the device to avoid people from accessing their router and get information out of it such as Wifi Password and all. Here is the full password configuration. I used simple Password for each device which I chose "cisco" to enter the Privileged EXEC Mode and "class" to enable and go to configuration mode.

- enable secret class
- line console 0
- password cisco
- login
- line vty 0 4
- password cisco
- login
- service password-encryption

  the banner is a message saying that only authorized users that are allowed. This might scare the unauthorized user when they read it.
- banner motd $AuthorizedUsersOnly$!

13 Testing/Result

After spending sometime on this project Access list how to control one or more network area It's is finally done after so many tries and understanding. I am going to test it out and show the result out of it.

In previous section where I tested the network without any ACL configuration all pc's were communicating and the Employers were able to access the FTP server without any issues. So now let's see if it works.

13.1 Testing the communication between Managers and Employers.

Yes as you can see in the picture below because I allowed the router to let the employers ping the managers and no other device besides them. By this configuration which allowed the communication **"permit ip 10.10.10.0 0.0.0.255 20.20.20.0 0.0.0.255**.

```
Pinging 20.20.20.10 with 32 bytes of data:

Request timed out.
Reply from 20.20.20.10: bytes=32 time=1ms TTL=127
Reply from 20.20.20.10: bytes=32 time<1ms TTL=127
Reply from 20.20.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 20.20.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

13.2 Testing if the Employers can ping and access the FTP.

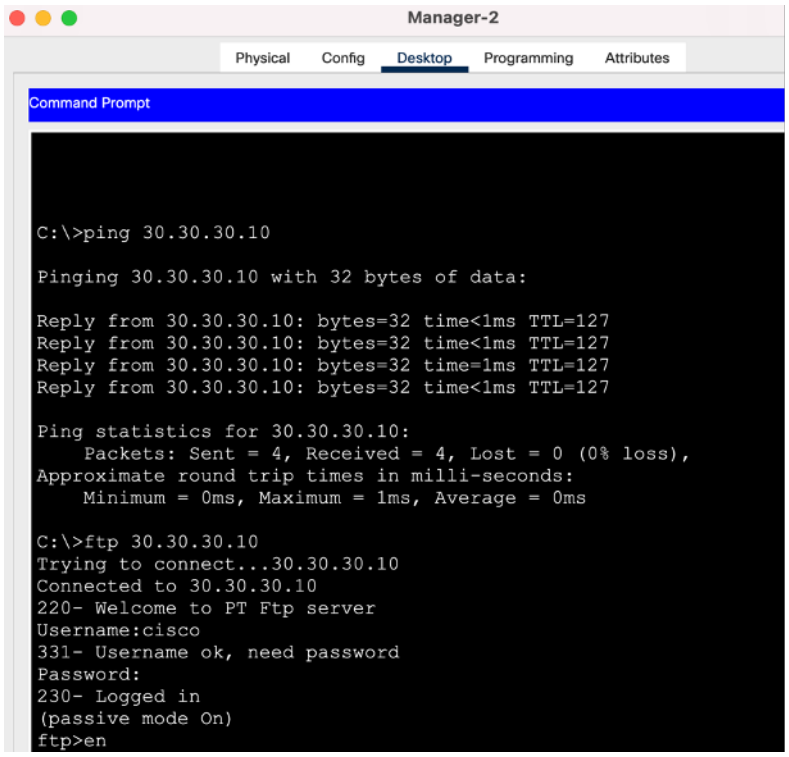
No the Employers cannot ping neither Access the FTP server due to the configuration I specified to not allow the communication or access the FTP **"deny ip 10.10.10.0 0.0.0.255 host 30.30.30.10" and "Deny udp 10.10.10.0 0.0.0.255 host 30.30.30.10 eq domain"**. However The employers can access the Webserver by **"permit tcp 10.10.10.0 0.0.0.255 host 30.30.30.10 eq www"** Configuration.

```
C:\>ping 30.30.30.10

Pinging 30.30.30.10 with 32 bytes of data:

Reply from 10.10.10.1: Destination host unreachable.
Reply from 10.10.10.1: Destination host unreachable.
Reply from 10.10.10.1: Destination host unreachable.
Reply from 10.10.10.1: Destination host unreachable.

Ping statistics for 30.30.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ftp 30.30.30.10
Trying to connect...30.30.30.10

%Error opening ftp://30.30.30.10/ (Timed out)
.



(Disconnecting from ftp server)
```

13.3 Testing the Manager network.

The Managers network should be able to access and communicate with the ftp server. Yes they have access to the Webserver and the FTP server. The server can receive pings from The managers.

Note that I haven't applied and ACL for manager network because you only I have specified which network are allowed to go through the router and which one can do what. For example if a different User from outside this network. they will not be able to go through because I permitted 10.10.10.0 and 20.20.20.20 only. ——————————-



13.4 Lastly this how it should look after apply password to each device.

The network looks more secure now, Unlike before the ACL configuration and password is being applied.

```
User Access Verification

Password:

R1>en
Password:
R1#
```

```
User Access Verification

Password:

SW1>en
Password:
SW1#
```

14 Problems Faced

During this project I have faced many issues even though it is for a small network. But the decision making where to implement the configuration and to what interface it should be applied to. Also the configurations were a bit confusing for me at first so I had to some research about it and understand it carefully what each configuration do and I got to understand it more eventhough some source's on the internet is hard to understand so in my 2 modules that I am studying "Automation and Disaster Recovery" and one of these modules lecturer was my supervisior MR.Usama. I've had some issue's with the IP addresses, sometimes I used 10.10.10.1 instead of 10.10.10.0. That's why I started writing the codes on a Notepad so that later on I can change the codes and verify it again to use it. In the end I made it and got it to work perfectly.

15 Conclusion and Future work.

15.1 Conclusion:

The project has been done successfully after many tries. The configuration that has been done on the router worked perfectly were it gave instructions to the devices for each network. Each network has it own rules, What can/cannot do.

Above all I managed to meet the project Objectives.

- Control the devices by setting rules

- Denying unnecessary devices to access the network

16 Future work

There are many idea's on how to improve the network. In my case there are some configuration that Packet tracer do not accept such as

- IPSEC a well password configuration maybe one of the best password security out there or maybe Apply AAA Configuration.
- Apply ACL with COPP "Control plane Policies" to troubleshoot the network such as limiting the traffic and for example you want to block ICMP echo request but want the devices to be able to get updated.
- Apply NAT AND PAT for further security if you are using acl only
- Assign Firewall to each device
- IPV6 Source guard: it blocks traffic that are coming from unknown ports.

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